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Bone Tumor Detection using X-Ray Images

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***Abstract*— Early and correct diagnosis of bone tumors is critical for successful diagnosis and treatment planning. The objective of this study is to propose and compare a robust diagnostic model for bone tumor detection based on X-ray images with multiple machine learning and deep learning models. Methods: Comparative analysis of four models of classification—Convolutional Neural Network (CNN), Stochastic Gradient Descent (SDG) Classifier, XGBoost, and ResNet-50—is done on the basis of their performance in bone tumor classification. The models are evaluated on key metrics like accuracy, precision, recall. Implementation: For enhancing clinical workflow usability, a GUI is designed using Tkinter that is user-friendly. The GUI facilitates input of X-ray images, to visually present model predictions, display segmentation output, and detect boundaries of tumors in real-time. Impact: Not only does the proposed framework detect the best classification model to identify bone tumors but also presents a combined system that can be used to support radiologists and clinicians in arriving at quicker and accurate diagnosis decisions.**

Keywords— CNN, SDG, XGBoost, Resnet-50, X-ray images, Bone tumor detection

I. INTRODUCTION

Bone tumors are any abnormal growth of tissue in or on a bone. Tumors can be benign, which means they are not cancerous and do not typically spread to other parts of the body, or malignant, which means that they are cancerous. It is important to diagnose them early and correctly, so tools can be provided to treat them effectively. The X-ray, CT, MRI diagnosis is time-consuming and may be subjective as well as error-prone because all these require manual reading by radiologists. AI and ML have been found useful to automate tumor detection and classification using medical imaging for accurate and efficient diagnosis. 10% - "Non-invasive surgery" is a surgery where no incision is made into the skin, most often used in orthopedic or spinal issues. An AI-based technique is used for medical imaging at diagnosis like detection and classification of tumors and all. This research paper proposes a diagnosis by computer-aided (CAD) system of the tumor detection using X-ray modality based on deep learning. The proposed system follows a structured workflow:

Image Pre-processing - converts image to grayscale, applies Gaussian filter for noise reduction, and enhances tumor region using Edge Detection Techniques. Classification – Implements algorithms like Stochastic Gradient Descent (SDG), CNN, XGBoost and Resnet-50 for classifying tumors as benign or malignant based on extracted features from pre-processed image. In segmentation we will get the tumor region only which will help us to see more clearly. A GUI using Tkinter for live image testing and classification. By using CNN, SDG, XGBoost and Resnet-50 models to detect bone tumor, this system aims to enhance diagnostic accuracy, improve model robustness, and reduce computational complexity.The proposed method will help to automate the tumor detection process so that it doesn't take too long and improve overall efficiency.

Flow of work is shown in figure 1. The pre-processed dataset is uploaded to the system, which is trained and validated by CNN, SDG, XGBoost and Resnet-50 algorithms and later has image uploaded for testing. Following this we will be seeing the output of classification as tumor, normal, as well as segmented image and an edge detected image.

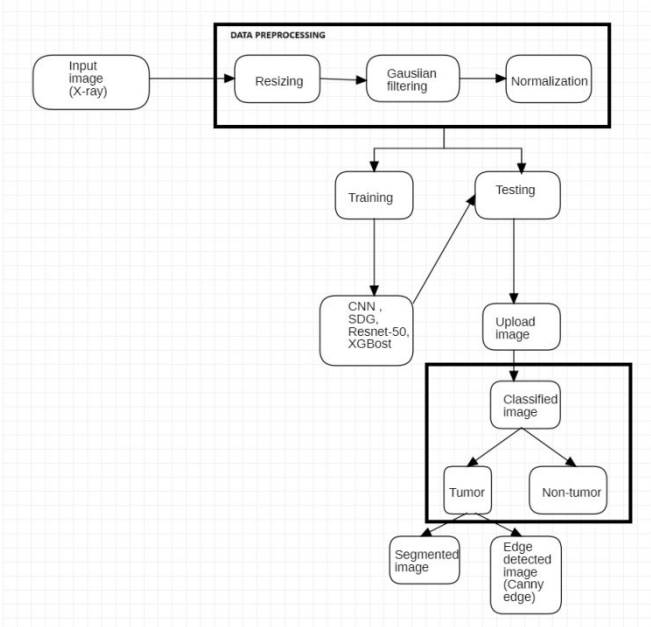

Figure 1 : Architecture Diagram of Proposed System

II. RELATED WORK

Bone tumor classification has been widely investigated using traditional ML methods like SVM, clustering algorithms, and others. The K-means clustering process implemented by Kadhim and Abdoon [1] extracts the bone tumor from CT and MRI images. It efficiently isolates the tumor area and enhances the classification performances An SVM-based classification technique used for bone cancer detection done by Shrivastava et al. [2] shows us the importance of feature extraction in increasing the accuracy. In their work, Krishnamoorthy and Madhurasree [9] proposed a feature extraction based machine learning approach showing that all the machine learning models are not required to depend heavily on advanced models. Instead, one can still get competitive results for classification, if the traditional models are used with advanced pre-processing. As deep learning progressed, CNN became a popular technique in the

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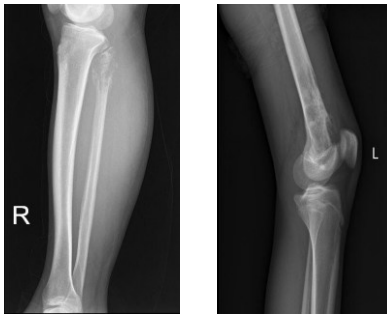
automated detection of tumors. Sampath et al. [3] performed a comparative study of CNN architectures based on CT images for the early diagnosis of bone cancer. Their study revealed that while traditional machine learning techniques are important, CNN-based models are more accurate. Do et al. [4] offers a Multilevel Seg-Unet Model for detection of knee bone tumor using global X-ray films and patch-based X-ray modality. It shows better accuracy in segmentation and classification. Furthermore, Aarthy et al. [5] proposed a Generative Deep Belief Neural Network (GDBNN) for multi-class bone cancer detection which employs deep learning to extract features and classify cancers. Their study showed better accuracy in classifying non-tumor, non-viable and viable tumors. Scientists are trying to use both conventional machine learning and deep learning together for hybrid models, to get better classification performance. An MRI based bone tumor detection system was proposed by Ambalkar and Thorat [6], which integrates feature extraction techniques with deep learning classifiers to provide robust performance on various datasets. The group of Gawade et al. [7] recommended the utilization of supervised deep-learning approaches for the observation of osteosarcoma with the help of ResNet and CNN classifiers for proper classification. Additionally, Singh and Vasumathi [8] presented a detailed review of deep learning techniques for segmentation and classification of bone tumours. The work emphasized issues related to data retrieval, preprocessing and annotation, and proposed solutions for deep learning models in medical images. Liu and Mongolia [10] expand the study by using machine learning models to identify bone metastasis among people with prostate cancer, showing the potential of predictive models. Apart from this the writing at the end. The conclusion is the literature says that the deep learning models if CNN based architecture outperform the conventional machine learning techniques in the classification of bone tumours. Yet, hybrid models that combine the two approaches also yield favorable results.

METHODOLOGY

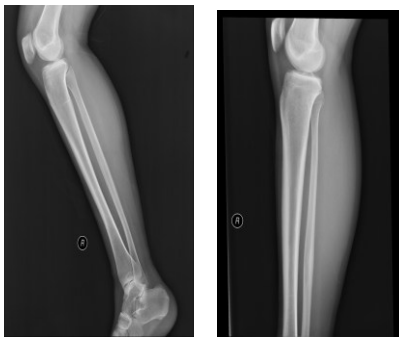
This model demonstrates a deep network architecture in the detection of bone tumor through CNN within the realm of deep learning. CNN, SDG, XGBoost and Resnet-50 is utilized for training a dataset.

A. Dataset

This experiment used a dataset downloaded from Figshare. Totally 2001 images out of which 1000 tumor and 1001 non-tumor images were used and pre-processed. The X-ray Images of bones were labelled as yes and no; the dimensions of the images are 668 x 1444 pixels. Figure 2 shows the sample X-ray images. 80% of the dataset were used for training purpose and 20% for testing purpose.



a) b)



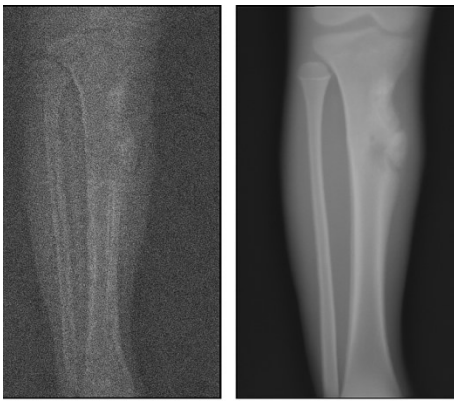
c) d)

Figure 2 : a) ,b) Tumor X-ray images ,
c) ,d) Non-tumor X-ray images

A. Preprocessing

The X-ray images are preprocessed to enhance relevant features and refine their quality. The preprocessing procedures are:

Image Acquisition: image is retrieved from Figshare, google 's dataset hub. Resizing: To provide consistent input dimensions throughout the dataset, input X-ray images are resized prior to processing. For segmentation, images are resized to 64×64 pixels, such that the deep learning model processes consistent input size. For classification, the images are resized to a consistent dimension depending on the model architecture (CNN, ResNet-50, XGBoost, or SDG classifier). The resizing is performed in a way that image size differences cannot affect model performance and assist in enabling improved generalization. Noise Reduction: Gaussian filtering is applied to remove noise and enhance image quality before segmentation and edge detection. Smoothing is done in a way that retains useful tumor features but erases small-scale intensity variations leading to false edges or improper segmentation. With the elimination of high-frequency noise, Gaussian filtering enhances boundary definition of tumors, leading to enhanced segmentation and classification. Applying Gaussian filtering to reduce noise and artifacts as shown in figure 3.



a) b)

Figure 3: a) X-ray image of leg bone
b) Noise reduced X-ray image

Normalization: Normalization helps all X-ray images have the same distribution of intensity which stops the model

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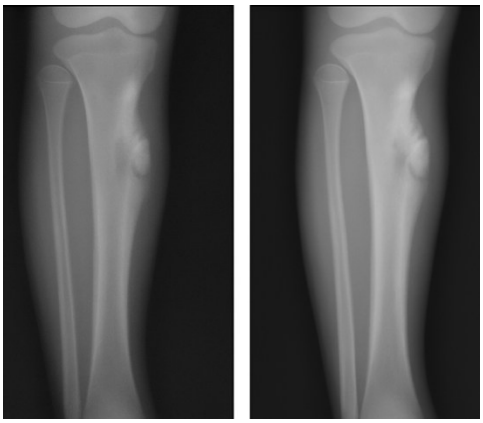
from being biased towards either a bright X-ray or a dark X-ray as shown in figure 4.



a) b)

Figure 4: a) X-ray image of leg bone ,
b) Normalized X-ray image

After all of these pre-processing techniques we will end up with a final X-ray image that would be easy to load and train it. The final processed image would look as shown in figure 5.

Figure 5 : Final X-ray image after pre-processing

B. Convolutional Neural Network (CNN)

CNNs proved to be reliable in carrying out the tasks of image recognition, therefore, making it the best equipment to analyze images of sign language. The CNN building blocks can be classified into convolutional, pooling, and fully connected layers. Convolution operations on input images help identify and detect features of varied types of thanks to these layers. Between the convolutional layers, pooling layers have been located at strategic points, more or less. Pooling or sub-sampling layers effectively reduce the spatial dimensions of the input while maintaining essential features. It really helps to reduce computational complexity and also the over-fitting chances. At the end of the CNN architecture, we connect layers that begin processing the information extracted by the convolutional layers to ultimately predict and classify whether a stroke is present. In the end, the flattening operation converts the 2D feature maps into 1D feature vectors. So, the data can be prepared to be fed into the fully connected layers. Finally, at the end of the CNN architecture, the fully connected layers work on the flattened feature vectors and produce stroke class scores. It has been designed such that all the neurons in the fully connected layers are connected to the neurons in the previous layer. The purpose of this is to fuse all the features obtained in the network. It's the CNN layer that leads to the final output layer; the purpose of the CNN layer is to combine information from earlier layers to produce a unique output.

C. SDG

A Stochastic Dual Gradient Descent (SDG) classifier was utilized to detect bone tumors, serving as a feature extractor and also as a classification model. In contrast to deep learning-based methods that leverage convolutional feature maps, SDG extracts features directly from the X-ray data for discriminating tumor and non-tumor regions. The classifier also does image segmentation and marks the affected regions, making it more interpretable for medical inspection. The comparison gives an idea about the compromises between machine learning techniques and deep learning methods for the detection of bone tumors.

D. Resnet-50

This study uses ResNet-50, one of 4 models – stochastic gradient descent (SDG), XGBoost and convolutional neural networks (CNN) to detect a bone tumor in X-ray images, alongside another two models. ResNet-50 is one of the deep learning architectures which use residual learning to make learning easy for deeper networks, therefore preventing the vanishing gradient issue. The 50-layers of convolutional layers with skip connections architecture of ResNet-50 amplify feature extraction and enhance classification accuracy. The case is different with ResNet-50 which learns features from the X-ray itself unlike SDG that take extracted features as input. There's no overlap and hence gives very good results between SDG and this. Since it is trained and tested independently to make the other classifiers fair, the CNN is not useful in feature extraction for this model

E. XGBoost

The Extreme Gradient Boosting (XGBoost) classifier was employed as a machine learning-based approach for the detection of bone tumors. XGBoost is highly known to be efficient, scalable, and perform well with structured data. In this work, XGBoost was used to train the extracted features of X-ray images to detect tumor and non-tumor cases. Contrary to deep neural networks that draw significant amounts of computational power, XGBoost efficiently handles high-dimensional data and maintains high classification accuracy. XGBoost brings together tree-based learning with gradient boosting in such a way that it learns high-order feature interactions and does not over-fit via L1 and L2 regularization techniques. Its parallelized nature also speeds up training, making it efficient for large datasets like the one employed in this research.

III. EXPERIMENTATION AND RESULTS

This section delves into the experimentation process and the obtained results of the CNN, SDG, XGBoost and Resnet-50 models for bone tumor detection.

Loading the dataset: The processed dataset is loaded into 2 classes. X-ray images are loaded into 2 as No tumor detected and tumor detected. Deep learning model: In deep learning, CNN-sequential model, SDG, XGBoost and Resnet-50 were used and total param and trainable params are calculated.

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Training: the dataset is trained into 15 epochs for Resnet-50 and CNN.

Segmentation: The bone tumor segmentation in this work is done through a deep learning-based method. A CNN-based segmentation model is helpful to identify the tumor area from X-ray images. The input image is resized to 64×64 pixels first to ensure consistency throughout the dataset. It is subsequently normalized to the range of [-1,1] for enhanced model performance. The resulting mask is resized to 300×300 pixels for improved visualization. The tumor is then Canny Edge Detection-processed to improve the tumor boundary. This deep learning-driven segmentation technique guarantees accurate tumor localization for classification and diagnostic evaluation. The region of the presence of tumor is separately segmented and shown as an image as shown in figure 6.

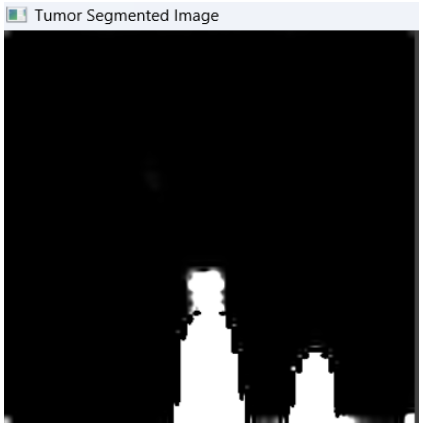


Figure 6 : Segmented image

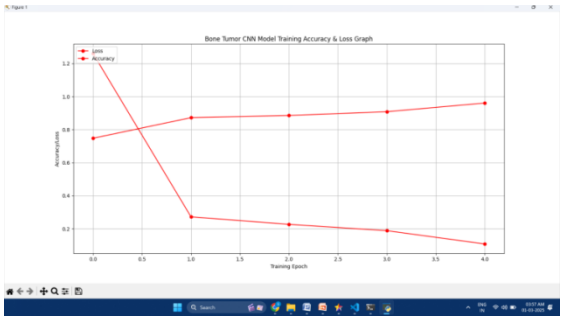
Edge detectetion: Canny Edge Detection is applied to extract the tumor edge from the segmented image in order to have improved visualization in X-ray scans. The process begins with Gaussian filtering to remove noise, and gradient computation employing Sobel operators to detect intensity changes. Non-maximum suppression is then used to keep only the most important edges. A double thresholding method categorizes edges as weak or strong, and the weak edges are only kept if they are linked to strong ones. Lastly, edge tracking by hysteresis ensures a smooth, defined boundary for the tumor. This method improves the accuracy of segmentation, giving a clearer image of the tumor to analyze later. The region where the tumor is present is marked in the uploaded X-ray image as shown in figure 7.

Figure 7: Edge detected image using Canny edge detection

Classification: Bone tumor classification in this paper is done using several algorithms architectures to identify between tumor and non-tumor images. The X-ray input image is resized initially to a normalized size to preserve consistency in the dataset. Normalization and other preprocessing methods are used to improve model performance. The preprocessed image is then passed through the trained classification model, which may be a Convolutional Neural Network (CNN), ResNet-50, XGBoost, or Stochastic Dual Gradient (SDG) classifier, based on the method employed. The model processes the image features and provides a classification label as to whether the input X-ray has a tumor or not. This classification process is vital for automatic diagnosis and helps in early bone tumor detection. Using CNN, SDG, XGBoost and Resnet-50 models it gives classified image into tumor and normal as shown in figure 8.

Figure 8: Classified image

Model performance: The validation accuracy and the validation loss is graphed as shown in figure 9 .



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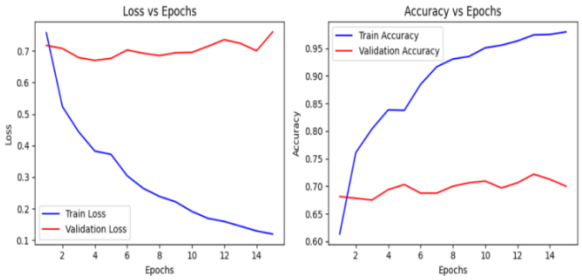

b)

Figure 9: a) CNN Accuracy and Loss Graph

b) Resnet-50 Accuracy and Loss Graph

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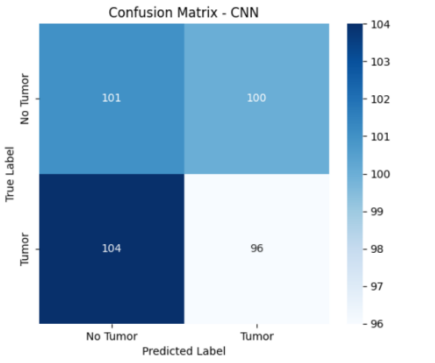

Figure 10: CNN confusion matrix

The performance metrics such as accuracy, recall and precision were used to evaluate the system performance. For accuracy equation (1) is used, for precision equation (2) is used and for recall equation (3) is used.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad \text{------} \rightarrow (1)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad \text{------} \rightarrow (2)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad \text{------} \rightarrow (3)$$

Figure 10 shows the confusion matrix for CNN model. This model provides the accuracy of 49.1%, recall of 48% and precision of 49%.

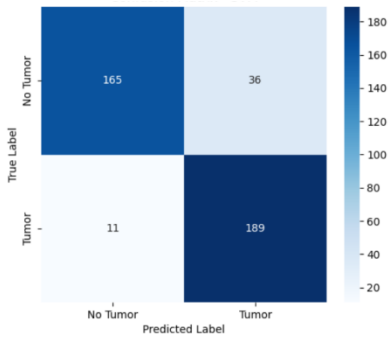

Figure 11: SDG confusion matrix

Figure 11 shows the confusion matrix for the SDG model. This model provides the accuracy 88.3%, recall 94.5%, precision 84%.

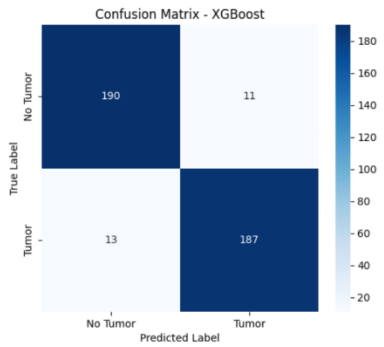

Figure 12: XGBoost confusion matrix

Figure 12 shows the confusion matrix for XGBoost model. This model provides the accuracy of 94%, recall of 93.5%, and precision of 94.4%.

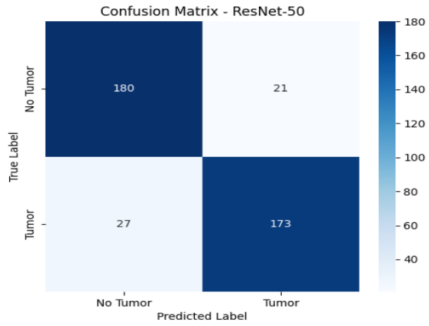

Figure 13: Resnet-50 confusion matrix

Figure 13 shows the confusion matrix for Resnet-50 model. This model provides the accuracy of 88%, recall 86.5%, precision 89.2%. The table 1 compares the performance of the CNN, SDG, XGBoost and Resnet-50 models. Table 1 evident that a XGBoost model has a higher accuracy of 94% than other models. From table 2, is clear that the proposed system performs well when compared to the existing system.

Table 1: Performance of the Proposed Models

MODEL	RECALL	PRECISION	ACCURACY	AUC
CNN	48	49	49.10%	0.45
SDG	94.5	84	88.30%	0.64
XGBoost	**93.5**	**94.4**	**94%**	**0.72**
Resnet 50	86.5	89.2	88%	0.81

Table 2: Performance comparison between proposed method and existing methods

Ref / Year	Technology Used	Accuracy (%)
[2] 2023	R-CNN	96
[7] 2024	Resnet-100	84
[4] 2024	Densenet	94
[6] 2024	CNN	96.7
[11] 2024	GCNN	80
[12] 2023	Resnet101	90.36
[14] 2021	CNN U-net	84.84
[18] 2024	Resnet152	97.7
Proposed Method	**XGBoost**	**94**

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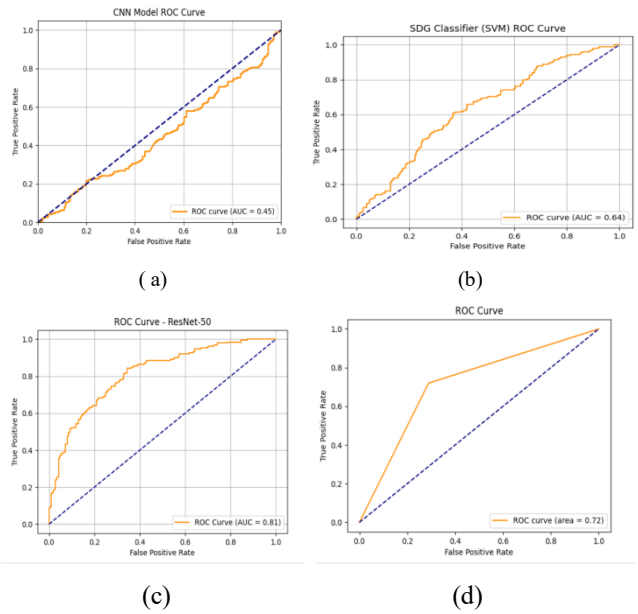

Figure 14: ROC curve of (a) CNN (b) SDG (c) Resnet50 (d) XGBoost

Figure 14 shows the ROC curve used to assess the effectiveness of four models and the Area under the curves value of CNN is 0.45, SDG 0.64, XGBoost 0.72, Resnet-50 0.81.

IV. CONCLUSION

In summary, this study suggests a complete and automated system for bone tumor identification from X-ray modality using both machine learning and deep learning methods. The experimental results indicate that the XGBoost classifier performed better than other models with a significant accuracy of 94%, which is a good candidate for bone tumor classification. The combination of segmentation and edge detection also adds to the interpretability and diagnostic value of the system. Moreover, the creation of an easy-to-use GUI enables real-time interaction and visualization, further increasing its potential clinical use.

There are some practical limitations that need to be considered. The dataset, although increased, can still be limited in diversity across patient populations, imaging conditions, and tumor types, which might limit model generalization in clinical practice. Downsampling of the image (e.g., 64×64 segmentation) can be lossy in terms of the important tumor information, impacting the localization accuracy. Additionally, while XGBoost is good at what it does, it lacks deep feature intuition in comparison with convolutional-based models.

Improvement on these disadvantages in the course of future efforts—using better-resolution inputs, including cutting-edge segmentation algorithms like U-Net, and with the use of clinically varied testing datasets—should assist in changing this system into a deployable clinical diagnostic application from being purely a research-grade prototype.

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